

REPORT

Title

Prolog Program to Implement Family Tree

Aim

To implement a **family tree using facts and rules in Prolog** and determine relationships between family members.

Problem Statement

Create a Prolog program that represents family members and their relationships. The program should determine relationships such as parent, father, mother, sibling, and grandparent.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp7.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp7.pl`.
3. Enter the family relationship facts and rules.
4. Save the program.
5. Load the file using `[exp7]..`
6. Execute queries to find different family relationships.
7. Observe the results.

Program

```
% Parent facts
parent(john, alice).
parent(mary, alice).
parent(john, bob).
parent(mary, bob).
parent(bob, charlie).
parent(susan, charlie).
```

```
% Father rule
```

```
father(X, Y) :-  
    parent(X, Y),  
    male(X).  
  
% Mother rule  
mother(X, Y) :-  
    parent(X, Y),  
    female(X).  
  
% Gender facts  
male(john).  
male(bob).  
male(charlie).  
  
female(mary).  
female(susan).  
female(alice).  
  
% Sibling rule  
sibling(X, Y) :-  
    parent(P, X),  
    parent(P, Y),  
    X \= Y.  
  
% Grandparent rule  
grandparent(X, Y) :-  
    parent(X, Z),  
    parent(Z, Y).
```

Sample Output

Query:

```
?- [exp7].  
true.
```

To find Alice's parent:

```
?- parent(X, alice).
```

Output:

```
X = john ;  
X = mary.
```

To find Alice's siblings:

```
?- sibling(alice, X).
```

Output:

```
X = bob.
```

To find Charlie's grandparents:

```
?- grandparent(X, charlie).
```

Output:

```
X = john ;
```

```
X = mary.
```

Conclusion

Thus, the **family tree was successfully implemented in Prolog** using facts and logical rules. The program can determine various relationships such as **parent, father, mother, sibling, and grandparent**.